

LAB OF ALGORITHM AND DATA STRUCRUTE

Quiz

1 : What is the time complexity of deleting an element from an array at a given position?

- A) $O(1)$
- B) $O(\log n)$
- C) $O(n)$
- D) $O(n \log n)$

Select Answer : C Correct

2 : Which of the following is NOT a necessary step when deleting an element from an array at a specified position?

- A) Finding the element's position.
- B) Shifting elements to fill the gap.
- C) Increasing the array's size.
- D) Updating the array size.

Select Answer : C Correct

3 : You have an array {10, 20, 30, 40, 50} and want to delete the element at position 2 (index 1). What will the array look like after deletion?

- A) {10, 20, 40, 50}
- B) {10, 30, 40, 50}
- C) {10, 20, 30, 50}
- D) {10, 40, 50}

Select Answer : B Correct

4 : What will happen if you try to delete an element from an array at a position that is out of bounds?

- A) The program will compile without errors.
- B) The program will crash.
- C) The element at the last position will be deleted.
- D) The program will print an error message.

Select Answer : B Correct

5 : What is the time complexity of a linear search in the worst case?

- A) $O(1)$
- B) $O(\log n)$
- C) $O(n)$
- D) $O(n \log n)$

Select Answer : C Correct

6 : Which of the following is a necessary condition for binary search to work correctly?

- A) The array must be sorted.
- B) The array must be unsorted.
- C) The array must contain only unique elements.
- D) The array must contain only even numbers.

Select Answer : A Correct

7 : In a linear search, if the target element is not found, what will be the return value (assuming we are returning the index)?

- A) -1

- B) 0
- C) The size of the array
- D) The middle index of the array

Select Answer : A Correct

8 : Binary search is significantly faster than linear search for:

- A) Small, unsorted arrays.
- B) Large, unsorted arrays.
- C) Large, sorted arrays.
- D) Small, sorted arrays.

Select Answer : C Correct

9 : Which loop is most commonly used to traverse an array in C++?

- A) for loop
- B) while loop
- C) do-while loop
- D) switch case

Select Answer : A Correct

10 : What is the time complexity of traversing an array of n elements using a for loop?

- A) $O(\log n)$
- B) $O(n)$
- C) $O(n^2)$
- D) $O(1)$

Select Answer : B Correct

11 : To insert an element at a specific position in an array, what must you do first (assuming the array is full)?

- A) Increase the array's size.
- B) Shift elements to the right of the insertion point.
- C) Overwrite the existing element at that position.
- D) Both A and B.

Select Answer : D Correct

12 : What will happen if you try to access an element at an index beyond the bounds of an array in C++?

- A) The program will automatically resize the array.
- B) The program will terminate with an error (e.g., segmentation fault).
- C) The program will continue to run but produce unexpected results.
- D) The program will ignore the access attempt.

Select Answer : B Correct